DentSAFER: Action-Aware Field Retrieval for Structured Dental CBCT Reporting

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Abstract. Dental cone-beam computed tomography (CBCT) provides volumetric evidence for diagnosis and treatment planning, but constructing a structured report from a full volume remains challenging because clinically relevant findings are distributed across numerous slices and must be consistently linked to patient-specific tooth positions, diagnoses, and treatment actions. This challenge is amplified under limited paired supervision, where unrestricted text generation is prone to memorizing training reports or producing unsupported clinical content. To address these challenges, we propose DentSAFER, a dental self-supervised action-aware field retrieval framework with entity refinement. DentSAFER first pretrains a compact 3D encoder using variance-invariance-covariance regularization and then jointly learns representations for seven report fields together with predictions for 32 FDI tooth positions and 21 treatment actions. At inference, the seven report fields are constructed through constrained retrieval from the labeled report library rather than unrestricted decoding. During retrieval, the diagnosis and treatment plan are selected jointly using action compatibility, while patient-specific tooth codes are removed from the retrieved text and replaced with tooth codes derived from image-conditioned predictions before final report assembly. On the official test set, DentSAFER achieved a Weighted Score of 0.2193, demonstrating the feasibility of structured dental report construction under limited paired supervision.

Keywords: Dental CBCT · Structured Reporting · Self-Supervised Learning · Multi-Task Learning · Field Retrieval

1 Introduction

Cone-beam computed tomography (CBCT) provides three-dimensional information about teeth, roots, alveolar bone, and adjacent maxillofacial structures, making it valuable for dental diagnosis and treatment planning. Constructing a structured report from an entire CBCT volume remains challenging because clinically relevant findings are distributed across numerous slices, and the target

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report must consistently integrate descriptive observations with patient-specific clinical information, including FDI tooth positions, diagnoses, and treatment actions. MMDental provides paired CBCT volumes and expert-authored clinical records for this task [11]. However, previous dental CBCT studies have focused primarily on tooth localization and semi-supervised segmentation [12,13,14], leaving structured report construction from full CBCT volumes comparatively underexplored.

In the broader field of medical report generation, existing studies have explored free-form decoding, retrieval-based prediction, and knowledge- or structure-guided modeling. Representative methods include memory-driven decoding [4], knowledge-guided retrieval and paraphrasing [8], contrastive report retrieval [5], fact-aware retrieval augmentation [9], and region- or anatomy-aware representation learning [10,15]. Recent volumetric systems have further extended medical image–text learning to three-dimensional CT data [1,6,7]. However, many of these methods are developed using substantially larger paired datasets or external pretrained models, exceeding the resources available in the present challenge setting. Under limited supervision, image-based representation pretraining offers a practical alternative. Transferable 3D representations have been explored by Med3D and Models Genesis [3,16], while VICReg provides a non-contrastive self-supervised objective that encourages invariant and non-collapsed representations [2].

The STS 2026 setting provides only 50 report-annotated CBCT volumes and 200 unlabeled volumes, making unrestricted text generation prone to memorizing the limited training reports and producing clinically unsupported content. We therefore propose DentSAFER, a self-supervised action-aware field retrieval framework with entity refinement that formulates structured dental reporting as constrained field retrieval. Specifically, a compact 3D encoder is first pretrained on the available CBCT images using VICReg and then jointly optimized to learn representations for seven report fields together with predictions for 32 FDI tooth positions and 21 treatment actions. At inference, DentSAFER combines field-wise retrieval with entity refinement to assemble the final report. Action compatibility guides retrieval of the diagnosis, treatment-plan, and handle fields, with the diagnosis and treatment plan sharing the same candidate report, whereas the remaining fields are retrieved independently. Patient-specific tooth codes are removed from retrieved text and replaced with tooth codes derived from image-conditioned predictions. This design leverages unlabeled CBCT data without relying on external pretrained weights, language models, or tooth segmentation and is designed to reduce cross-field inconsistency and the transfer of patient-specific tooth identifiers. The resulting framework achieved a Weighted Score of 0.2193 on the official test set.

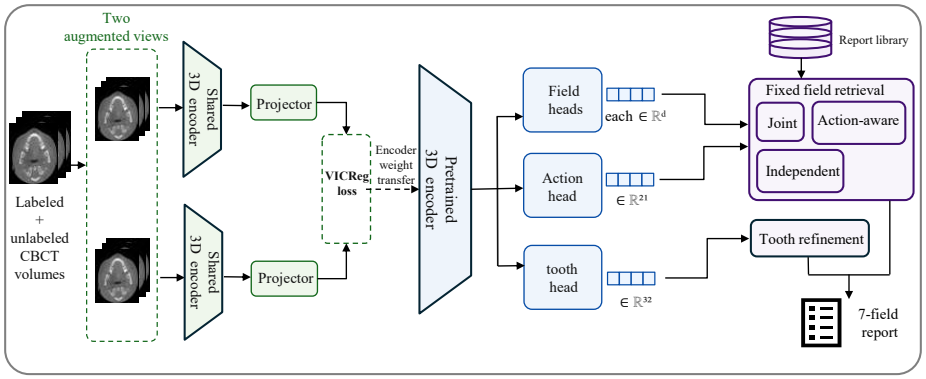


Fig. 1. Overview of DentSAFER. VICReg pretraining on labeled and unlabeled CBCT volumes provides encoder initialization for supervised multi-task learning of field embeddings, tooth-position probabilities, and treatment-action probabilities. At inference, action-aware field retrieval from the labeled report library is combined with image-conditioned tooth-code refinement to assemble the final seven-field report.

2 Method

2.1 Overview

Let $\mathcal{D}_L = \{(x_i, r_i)\}_{i=1}^{N_L}$ denote the labeled dataset, where x_i is a CBCT volume and $r_i = (r_{i,f})_{f\in\mathcal{F}}$ is its corresponding seven-field report, with $\mathcal{F} = \{1, \ldots, 7\}$ denoting the field index set. The $\mathcal{D}_U = \{x_j\}_{j=1}^{N_U}$ denote the unlabeled CBCT dataset, where $N_L = 50$ and $N_U = 200$. Following the official data schema, the seven report fields, ordered by their indices in $\mathcal{F}$, are "main appeal," "present medical history," "oral check," "diagnosis," "treatment plan," "handle," and "doctor advices." From each labeled report, we additionally derive a 32-dimensional multi-label FDI tooth-position target $y_i^{\text{tooth}} \in \{0,1\}^{32}$ and a 21-dimensional multi-label treatment-action target $y_i^{\text{act}} \in \{0,1\}^{21}$.

Figure 1 illustrates the overall pipeline of DentSAFER. The framework first performs VICReg-based self-supervised pretraining on the labeled and unlabeled CBCT volumes, and the learned encoder weights are subsequently transferred to the supervised multi-task model. The supervised stage then jointly predicts seven report-field embeddings, 32-dimensional FDI tooth-position logits, and 21-dimensional treatment-action logits. At inference, the predicted field embeddings and treatment-action probabilities guide constrained, action-aware retrieval from the labeled report library, while tooth-position probabilities support entity refinement before final report assembly.

2.2 Preprocessing and Augmentation

Both compressed and uncompressed NIfTI-1 files are loaded, with voxel intensities adjusted using the stored slope and intercept. We uniformly sample 128, 128,

and 80 indices along the original X, Y, and Z axes, respectively, and transpose the resulting array to a fixed input size of $80 \times 128 \times 128$. Finite voxel intensities are clipped to the 0.5th and 99.5th percentiles of each volume and then scaled to $[-1, 1]$. The same preprocessing pipeline is used during training and inference. No segmentation, registration, manual region selection, or spacing-aware physical-space resampling is performed.

During training, intensity gain and offset are applied with probability 0.9 using $g \sim \mathcal{U}(0.85, 1.15)$ and $b \sim \mathcal{U}(-0.12, 0.12)$. Gaussian noise is added with probability 0.7, with its standard deviation sampled as $\sigma \sim \mathcal{U}(0.005, 0.04)$. Random translation of up to three voxels along the depth axis and five pixels along the in-plane axes is applied with probability 0.7. A randomly located cuboid region is masked with probability 0.5.

2.3 Self-Supervised 3D Encoder

The encoder comprises a 3D convolutional stem with a $5 \times 7 \times 7$ kernel, a max-pooling layer, and four residual stages. The four stages use 24, 48, 96, and 192 channels, respectively. After the final residual stage, global average pooling followed by a linear layer produces a 256-dimensional volume representation. A three-layer projector then maps this representation to a 128-dimensional feature used for VICReg pretraining.

The encoder and projector process two independently augmented views of each CBCT volume and are jointly optimized using the VICReg objective:

$$\mathcal{L}_{\mathrm{SSL}} = 25\mathcal{L}_{\mathrm{inv}} + 25\mathcal{L}_{\mathrm{var}} + \mathcal{L}_{\mathrm{cov}}, \tag{1}$$

where the invariance, variance, and covariance terms encourage cross-view agreement, prevent representation collapse, and decorrelate feature dimensions, respectively [2].

2.4 Multi-Task Field and Entity Learning

For supervised training, the VICReg projector is discarded, while the pretrained encoder is retained and fine-tuned jointly with the task-specific heads. Let h_i denote the encoder representation of volume x_i. Two linear classification heads map h_i to 32-dimensional FDI tooth-position logits and 21-dimensional treatment-action logits. The same encoder representation is passed through a shared LayerNorm–Linear–SiLU–Dropout block followed by seven field-specific linear heads. For each field $f \in \mathcal{F}$, the corresponding head produces a 256-dimensional prediction $p_{i,f}$. To construct field-level text targets without an external text encoder, each field is lowercased and represented using word-level 1–3-grams and within-token character-level 3–5-grams. The resulting target representation for field f of report i is denoted by $t_{i,f}$.

Let $m_{i,f}$ denote the validity mask for field f and w_f denote its field-specific weight. Given $p_{i,f}$ and $t_{i,f}$, we define the field-representation loss as a weighted

cosine-distance objective:

$$\mathcal{L}_{\text{field}} = \frac{\sum_{i,f} m_{if} w_f (1 - p_{if}^{\top} t_{if})}{\sum_{i,f} m_{if} w_f}, \tag{2}$$

where field weights are $w_f = (0.10, 0.11, 0.18, 0.20, 0.18, 0.13, 0.10)$. The FDI tooth-position and treatment-action heads are optimized using class-weighted binary cross-entropy losses, $\mathcal{L}_{\text{tooth}}$ and $\mathcal{L}_{\text{act}}$, respectively. For supervised training, positive-class weights are estimated for the tooth-position and treatment-action labels and clipped to $[1, 8]$. The overall supervised objective is:

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{field}} + 0.35\mathcal{L}_{\text{tooth}} + 0.20\mathcal{L}_{\text{act}}. \tag{3}$$

Within each supervised fold, the held-out CBCT–report pairs are excluded from supervised parameter updates. Their report texts are excluded from text-feature fitting and retrieval-library construction, while the derived tooth-position and treatment-action targets are excluded from supervised loss computation.

2.5 Action-Aware Field Retrieval and Entity Refinement

For supervised model m, let $p_f^{(m)}$ denote the predicted representation of field f, and let $t_{if}^{(m)}$ denote the corresponding representation of field f in candidate report i. Their cosine similarity is

$$s_{if}^{(m)} = \left(t_{if}^{(m)}\right)^{\top} p_f^{(m)}. \tag{4}$$

Let $\mathcal{M}_i$ denote the set of supervised models for which candidate report i is included in the corresponding training retrieval library. The cross-model similarity for candidate i and field f is aggregated as

$$\bar{s}_{if} = \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} s_{if}^{(m)}. \tag{5}$$

Each labeled report is held out in one fold and appears in the retrieval libraries of the remaining four models; therefore, $|\mathcal{M}_i| = 4$. In contrast, tooth and action probabilities are averaged across all five models:

$$\bar{u} = \frac{1}{5} \sum_{m=1}^{5} u^{(m)}, \qquad \bar{a} = \frac{1}{5} \sum_{m=1}^{5} a^{(m)}, \tag{6}$$

where $\bar{u} \in [0, 1]^{32}$ and $\bar{a} \in [0, 1]^{21}$ denote the averaged tooth and action probabilities, respectively.

For the "main appeal," "present medical history," "oral check," and "doctor advices" fields, the highest-scoring nonempty candidate is retrieved independently by maximizing $\bar{s}_{i,f}$. The "diagnosis" and "treatment plan" fields are instead retrieved jointly from a shared candidate report using both field similarity

and treatment-action compatibility. Given averaged action probabilities $\bar{a}$ and candidate action target q_i, action compatibility is defined as:

$$c_i = \frac{2q_i^\top \bar{a}}{\sum_k q_{ik} + \sum_k \bar{a}_k}. \tag{7}$$

The shared candidate for the diagnosis and treatment-plan fields is selected as:

$$i^* = \arg\max_i \left(0.70\bar{s}_{i,\text{diag}} + 0.30\bar{s}_{i,\text{plan}} + 0.30c_i\right). \tag{8}$$

For the "handle" field, the candidate is selected using the action-aware score $\bar{s}_{i,\text{handle}} + 0.30c_i$.

During entity refinement, tokens matching the predefined FDI tooth-code pattern are removed from all retrieved fields. The ensemble tooth-position probabilities $\bar{\mathbf{u}}$ are thresholded at 0.50 and ranked in descending order. At most five predicted FDI codes are retained and appended to the "oral check" field.

3 Experiments

3.1 Dataset Splits and Evaluation Metrics

The dataset contains 50 labeled and 200 unlabeled CBCT volumes. The 50 labeled cases are organized into five fixed folds, with 40 cases used for supervised multi-task training and 10 cases held out in each fold. Within each fold, field-specific text statistics, field-level text targets, class weights, and the retrieval library are constructed exclusively from the 40 training reports. The 10 held-out cases are used only for checkpoint selection and fold-wise validation.

Before supervised multi-task learning, a shared encoder is pretrained using VICReg on the 250 available CBCT volumes, with 225 volumes used for self-supervised training and 25 for validation. The pretrained encoder is subsequently used to initialize the supervised models. For official evaluation, the test volumes are processed only during inference using the frozen model ensemble. The official scorer evaluates textual similarity using BLEU-1–4, ROUGE-L, METEOR, and CIDEr; clinical entities using tooth, diagnosis, action, plan, and medication metrics; and report completeness using field-completion measures and an overall Weighted Score.

3.2 Implementation Details

All experiments were conducted on a workstation equipped with an NVIDIA RTX 4090 GPU. The implementation, optimization, computational cost, and runtime settings are summarized in Table 1.

Table 1. Implementation, training, and runtime configuration.

Item	Setting
Environment	Ubuntu 20.04; 12 CPU cores; 28 GB RAM; NVIDIA RTX 4090 (24 GB); CUDA 11.8; PyTorch 2.1.2
Input/batch	$80 \times 128 \times 128$; batch 8 (SSL), 4 (supervised)
Optimization	AdamW, weight decay 10^{-4}; LR 2×10^{-4} (SSL), $5 \times 10^{-5}/3 \times 10^{-4}$ (encoder/head)
Schedule	80 epochs; cosine annealing; mixed precision; gradient clipping 5.0
Model cost	5.180 M parameters (SSL), 5.258 M (supervised); ∼6.493 GFLOPs/model/volume
Training time	5 min 4 s (SSL) + 3 min 12 s (parallel supervised schedule); 8 min 16 s total
Inference time	114 s for 50 cases; 2.28 s/case

Table 2. Official test-set results. P, R, and F1 denote precision, recall, and F1-score.

Metric	Score	Metric	Score
Weighted Score	0.2193	Diagnosis P	0.0733
BLEU-1	0.1957	Diagnosis R	0.1540
BLEU-2	0.0735	Diagnosis F1	0.0930
BLEU-3	0.0434	Tooth P	0.0897
BLEU-4	0.0315	Tooth R	0.1912
ROUGE-L	0.2171	Tooth F1	0.1093
METEOR	0.2417	Action P	0.2693
CIDEr	0.0602	Action R	0.4137
Exact match	0.0200	Action F1	0.3094
Partial match	0.1340	Medication F1	0.6100
Plan recall	0.4737	Pred. complete	1.0000
GT complete	0.9000	Field complete	1.0000

4 Results and Discussion

4.1 Challenge Results

Table 2 summarizes the organizer-returned metrics. BLEU-1, ROUGE-L, and METEOR were higher than BLEU-4, indicating greater agreement at the token and partial-sequence levels than at the exact multi-token level. Medication F1 and treatment-plan recall were 0.6100 and 0.4737, respectively, whereas diagnosis and tooth-position F1 were 0.0930 and 0.1093.

Figure 2 presents qualitative comparisons of two cases. In Case 1, the predicted report captures dental caries and the associated filling and removal actions. In Case 2, it identifies impacted teeth and provides a corresponding extraction recommendation. Overall, these examples illustrate the ability of the predictions to capture clinically relevant diagnostic findings and generate appropriate treatment recommendations.

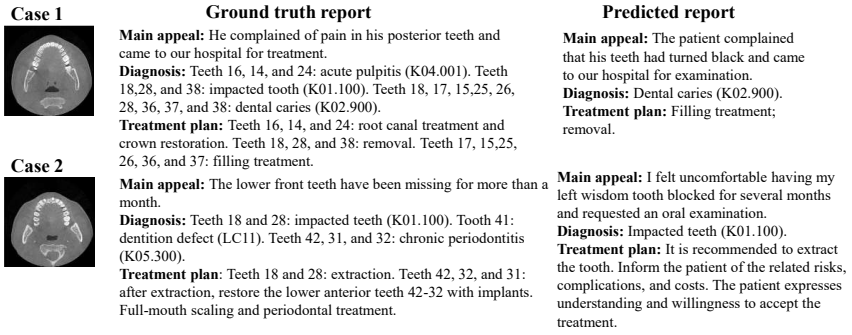


Fig. 2. Qualitative comparison of ground-truth and predicted reports. Each case includes a representative axial CBCT slice and the corresponding main appeal, diagnosis, and treatment plan.

4.2 Discussion and Limitations

With only 50 labeled CBCT volumes, unrestricted generation is prone to unstable or unsupported content. Field retrieval and entity refinement therefore prioritize structural completeness, producing all seven fields for every official case. However, the low diagnosis and tooth-position F1 scores indicate that complete output does not guarantee clinically accurate content. Qualitative results suggest that the model captures frequent findings and broad treatment actions but often under-specifies complex cases, omitting uncommon diagnoses, tooth positions, or case-specific procedures. This limitation is likely related to the small retrieval library, global volume representation, and fixed entity thresholds. The displayed axial slices provide anatomical context only and do not reveal the regions driving the predictions. The study also lacks controlled ablations, external validation, and clinician-adjudicated error analysis. Future work should explore tooth-local representations, constrained diagnosis–tooth–treatment linking, calibrated uncertainty, and larger clinician-verified retrieval libraries.

5 Conclusion

We present DentSAFER, a self-supervised action-aware field retrieval framework with entity refinement for seven-field dental report construction from CBCT volumes under limited paired supervision. The framework combines VICReg pretraining on labeled and unlabeled CBCT volumes with multi-task prediction of report-field embeddings, FDI tooth positions, and treatment actions, followed by constrained action-aware retrieval and image-conditioned tooth-code refinement. On the official test set, DentSAFER achieved a Weighted Score of 0.2193 and generated all seven report fields for every case. Diagnosis coding and tooth localization remain the main areas for future improvement.

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